



A NOTE ON INTERIOR C^2 ESTIMATE FOR GENERAL HESSIAN QUOTIENT EQUATION

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ABSTRACT. In this note, we study the interior C^2 regularity problem for the general Hessian quotient equation $\frac{\sigma_k}{\sigma_l}(D^2u) = f$. Under a concavity assumption, we obtain an interior C^2 estimate for convex solutions when $l = k-1, k-2$. It is known that such estimate fails when $l \leq k-3$.

1. Introduction. The study of interior C^2 estimate for fully nonlinear equations is of great interest due to its elegant form, namely the estimate does not depend on any information of the solution on the boundary. It can be traced back to Heinz's work on the Weyl embedding problem. In [9], Heinz established an interior C^2 estimate for the Monge-Ampère equation $\det(D^2u) = f$ for $n = 2$. See also recent proofs by Chen, Han and Ou [2] and Liu [12]. However, interior C^2 estimate fails for the Monge-Ampère equation when $n \geq 3$, as demonstrated by the counterexample of Pogorelov [17]. Building on this, Urbas [21] showed that interior C^2 estimate also fails for the Hessian equation $\sigma_k(D^2u) = f$ for $k \geq 3$. The remaining case is the σ_2 equation $\sigma_2(D^2u) = f$, which has attracted substantial attention.

For the σ_2 equation, a major breakthrough was made by Warren and Yuan. In [23], they obtained an interior C^2 estimate for $\sigma_2(D^2u) = 1$ for $n = 3$. For general right-hand side f in dimension 3, interior C^2 estimate was derived by Qiu [18]. In a cutting-edge paper [20], Shankar and Yuan established an interior C^2 estimate for $\sigma_2(D^2u) = 1$ in dimension 4. For general right-hand side f in dimension 4, interior C^2 estimate was recently proved by Fan [5] by following the method of Shankar and Yuan [20].

For general n , interior C^2 estimate was proved under various conditions. Interior C^2 estimate for $\sigma_2(D^2u) = 1$ was proved by McGonagle, Song and Yuan [15] under the condition $D^2u > -c(n)I$; by Shankar and Yuan [19] under the condition $D^2u > -KI$; by Shankar and Yuan [20] under the condition $D^2u \geq -c(n)\Delta u I$; and by Mooney [16] under the condition $D^2u > 0$. Interior C^2 estimate for $\sigma_2(D^2u) = f$ was proved by Guan and Qiu [6] under the condition $\sigma_3(D^2u) > -A$; and by Fan [5] under the condition $D^2u \geq -c(n)\Delta u I$. See also related work in [4].

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Another natural class of fully nonlinear equations is the Hessian quotient equation $\frac{\sigma_k}{\sigma_l}(D^2u) = f$. For $\frac{\sigma_3}{\sigma_1} = f$ in dimensions 3 and 4, interior C^2 estimate was obtained by Chen, Warren and Yuan [3], Wang and Yuan [22] and Zhou [25], by exploiting the special Lagrangian structure of the equation. In recent works [13, 14], the first author established an interior C^2 estimate for $\frac{\sigma_n}{\sigma_k}(D^2u) = f$ when $k = n-1, n-2$, by using Legendre transform. For the general Hessian quotient equation $\frac{\sigma_k}{\sigma_l}(D^2u) = f$, singular solution was constructed in [14] when $l \leq k-3$. The remaining case is $\frac{\sigma_k}{\sigma_l}(D^2u) = f$ when $l = k-1, k-2$.

The purpose of this paper is to prove the interior C^2 estimate in this case under the following natural concavity assumption.

Assumption 1.1. Let $n \geq 2$, $1 \leq l < k \leq n$ with $l = k-1, k-2$, and let $F = \frac{\sigma_k}{\sigma_l}$. Suppose $\lambda_1 \geq \dots \geq \lambda_n > 0$. Then for any $\xi \in \mathbb{R}^n$, there exist constants $K_0 \geq 1$ and $K_1 \geq 1$, depending only on n, k and F , such that whenever $\lambda_1 \geq K_0$, we have

$$-\sum_{i,j} F^{ii,jj} \xi_i \xi_j + \sum_{\lambda_i \neq \lambda_1} \frac{2F^{ii} \xi_i^2}{\lambda_1 - \lambda_i} + \frac{K_1}{F} \left(\sum_i F^{ii} \xi_i \right)^2 \geq (1 + c(n, k)) \frac{F^{11} \xi_1^2}{\lambda_1},$$

where $c(n, k)$ is a positive constant depending only on n and k . Here $F^{ii} = \frac{\partial F}{\partial \lambda_i}$ denotes the first derivative of F with respect to λ_i , and $F^{ii,jj} = \frac{\partial^2 F}{\partial \lambda_i \partial \lambda_j}$ denotes the second derivative of F with respect to λ_i and λ_j .

Under the concavity assumption, we obtain the following interior C^2 estimate.

Theorem 1.2. Let $n \geq 2$, $1 \leq l < k \leq n$ with $l = k-1, k-2$. Let $f \in C^{1,1}(B_{10} \times \mathbb{R})$ be a positive function and let $u \in C^4(B_{10})$ be a convex solution of

$$F(D^2u) = \frac{\sigma_k}{\sigma_l}(D^2u) = f(x, u), \quad \text{in } B_{10} \subset \mathbb{R}^n. \quad (1.1)$$

Assume that Assumption 1.1 holds. Then we have

$$|D^2u(0)| \leq C,$$

where C depends only on $n, k, \|u\|_{C^{0,1}(\overline{B}_9)}, \min_{\overline{B}_9 \times [-M, M]} f$ and $\|f\|_{C^{1,1}(\overline{B}_9 \times [-M, M])}$. Here M is a large constant satisfying $\|u\|_{L^\infty(\overline{B}_9)} \leq M$.

Remark 1.3. Assumption 1.1 arises naturally from the computation of the largest eigenvalue of D^2u . In fact, this concavity inequality is the key to various C^2 estimate for fully nonlinear equations. When $F = \sigma_k$, this concavity is known to hold for convex solutions, proved recently by Zhang [24].

Remark 1.4. We remark that for $k = n$, Assumption 1.1 was proved by Guan and Sroka [7]. As mentioned above, this concavity holds for Hessian operator [24]. Therefore, we believe it should also hold for Hessian quotient operator.

The paper is organized as follows. In Section 2, we collect basic properties of the Hessian and Hessian quotient operators and establish a Jacobi inequality. In Section 3, we perform the Legendre transform and derive an integral bound for b . Finally, in Section 4, we complete the proof of Theorem 1.2 using integration by parts.

2. Preliminaries. In this section, we collect some basic properties of the Hessian and Hessian quotient operators, and make a simple observation about our equation. We also state the Jacobi inequality.

Let $\lambda = (\lambda_1, \dots, \lambda_n) \in \mathbb{R}^n$. We denote

$$(\lambda|i) = (\lambda_1, \dots, \lambda_{i-1}, \lambda_{i+1}, \dots, \lambda_n) \in \mathbb{R}^{n-1},$$

that is, $(\lambda|i)$ is the vector obtained by deleting the i -th component of λ . Similarly, $(\lambda|ij)$ is the vector obtained by deleting the i -th and j -th components of λ .

We first collect some basic formulas for the Hessian operators; see for instance [11].

Lemma 2.1. For any $\lambda = (\lambda_1, \dots, \lambda_n) \in \mathbb{R}^n$, we have

$$\sigma_k(\lambda) = \lambda_i \sigma_{k-1}(\lambda|i) + \sigma_k(\lambda|i), \quad \sum_i \sigma_k(\lambda|i) = (n-k)\sigma_k(\lambda),$$

$$\sum_i \lambda_i \sigma_{k-1}(\lambda|i) = k\sigma_k(\lambda), \quad \sum_i \lambda_i^2 \sigma_{k-1}(\lambda|i) = \sigma_1(\lambda)\sigma_k(\lambda) - (k+1)\sigma_{k+1}(\lambda).$$

For simplicity, we write $\sigma_k(|i)$ for $\sigma_k(\lambda|i)$ when there is no ambiguity. Similarly, we write $\sigma_k(|ij)$ for $\sigma_k(\lambda|ij)$.

Let F be a symmetric function of λ , where $\lambda = (\lambda_1, \dots, \lambda_n)$ are the eigenvalues of D^2u . We extend F to a function of D^2u and denote

$$F^{pq} = \frac{\partial F}{\partial u_{pq}}, \quad F^{pq,rs} = \frac{\partial^2 F}{\partial u_{pq} \partial u_{rs}}.$$

We now collect some basic properties of the Hessian quotient operator.

Lemma 2.2. Let $n \geq 2$, $1 \leq l < k \leq n$, and let $F = \frac{\sigma_k}{\sigma_l}$. Suppose D^2u is diagonalized at x_0 . Then at x_0 we have

$$F^{pq} = \left(\frac{\sigma_k^{pq}}{\sigma_l} - \frac{\sigma_k \sigma_l^{pq}}{\sigma_l^2} \right) \delta_{pq},$$

$$F^{pq,rs} = \begin{cases} -2 \frac{\sigma_k^{pp} \sigma_l^{pp}}{\sigma_l^2} + 2 \frac{\sigma_k (\sigma_l^{pp})^2}{\sigma_l^3}, & p = q = r = s, \\ \frac{\sigma_k^{pp,rr}}{\sigma_l} - \frac{\sigma_k^{pp} \sigma_l^{rr}}{\sigma_l^2} - \frac{\sigma_k^{rr} \sigma_l^{pp}}{\sigma_l^2} - \frac{\sigma_k \sigma_l^{pp,rr}}{\sigma_l^2} + 2 \frac{\sigma_k \sigma_l^{pp} \sigma_l^{rr}}{\sigma_l^3}, & p = q, r = s, p \neq r, \\ \frac{\sigma_k^{pq,qp}}{\sigma_l} - \frac{\sigma_k \sigma_l^{pq,qp}}{\sigma_l^2}, & p = s, q = r, p \neq q, \\ 0, & \text{otherwise.} \end{cases}$$

In particular, for all $p \neq q$, we have

$$-F^{pq,qp} = \frac{F^{pp} - F^{qq}}{\lambda_q - \lambda_p}, \quad \lambda_p \neq \lambda_q,$$

where λ_p and λ_q are the eigenvalues of D^2u .

We also need the following observations on the Hessian quotient operator.

Lemma 2.3. Let $n \geq 2$, $1 < k \leq n$, and let $F = \frac{\sigma_k}{\sigma_{k-1}}$. Suppose $\lambda_1 \geq \dots \geq \lambda_n > 0$. Then we have

$$\frac{F}{c(n,k)} \leq \lambda_k \leq c(n,k)F,$$

where $c(n,k)$ is a positive constant depending only on n and k .

Proof. We have

$$\frac{\lambda_k}{c(n, k)} \leq F = \frac{\sigma_k}{\sigma_{k-1}} \leq c(n, k)\lambda_k.$$

It follows that

$$\frac{F}{c(n, k)} \leq \lambda_k \leq c(n, k)F.$$

□

Lemma 2.4. Let $n \geq 3$, $2 < k \leq n$, and let $F = \frac{\sigma_k}{\sigma_{k-2}}$. Suppose $\lambda_1 \geq \dots \geq \lambda_n > 0$. Then we have

$$\frac{F}{c(n, k)} \leq \lambda_{k-1}\lambda_k \leq c(n, k)F, \quad \lambda_{k-1} \geq \frac{\sqrt{F}}{c(n, k)}, \quad \lambda_k \leq c(n, k)\sqrt{F},$$

where $c(n, k)$ is a positive constant depending only on n and k .

Proof. We have

$$\frac{\lambda_k^2}{c(n, k)} \leq \frac{\lambda_{k-1}\lambda_k}{c(n, k)} \leq F = \frac{\sigma_k}{\sigma_{k-2}} \leq c(n, k)\lambda_{k-1}\lambda_k \leq c(n, k)\lambda_{k-1}^2.$$

It follows that

$$\frac{F}{c(n, k)} \leq \lambda_{k-1}\lambda_k \leq c(n, k)F, \quad \lambda_{k-1} \geq \frac{\sqrt{F}}{c(n, k)}, \quad \lambda_k \leq c(n, k)\sqrt{F}.$$

□

We now state the Jacobi inequality, which is a direct corollary of Assumption 1.1.

Lemma 2.5. Let $n \geq 2$, $1 \leq l < k \leq n$ with $l = k-1, k-2$. Let $f \in C^{1,1}(B_{10} \times \mathbb{R})$ be a positive function and let $u \in C^4(B_{10})$ be a convex solution of (1.1). For any $x_0 \in B_9$, suppose that D^2u is diagonalized at x_0 such that $\lambda_1 \geq \dots \geq \lambda_n$.

Assume that Assumption 1.1 holds. Suppose $\lambda_1 \geq K_0$, where K_0 is the constant in Assumption 1.1. Let $b = \ln \lambda_1$, then at x_0 , we have

$$\sum_i F^{ii}b_{ii} \geq c(n, k) \sum_i F^{ii}b_i^2 - C,$$

in the viscosity sense, where $c(n, k)$ is a positive constant depending only on n and k , and C depends only on $n, k, \|u\|_{C^{0,1}(\overline{B}_9)}, \min_{\overline{B}_9 \times [-M, M]} f$ and $\|f\|_{C^{1,1}(\overline{B}_9 \times [-M, M])}$. Here M is a large constant satisfying $\|u\|_{L^\infty(\overline{B}_9)} \leq M$.

Proof. The proof is almost the same as that of Lemma 4.1 in [14]. The only difference is that, instead of using Lemma 3.1 and Lemma 3.2 in [14], we use Assumption 1.1 here. The remaining steps are identical, so we omit the details. □

3. Legendre transform. In this section, we adopt the idea of Shankar and Yuan [19] and use the Legendre transform to bound b by its integral. The key observation is that, after the transformation, b becomes a subsolution of a uniformly elliptic equation.

Consider the Legendre transform of the function $u + \frac{1}{2}|x|^2$. We have

$$(x, Du(x) + x) = (Dw(y), y),$$

where $w(y)$ is the Legendre transform of $u + \frac{1}{2}|x|^2$. Note that $y(x) = Du(x) + x$ is a diffeomorphism.

Define

$$G(D^2w(y)) = -F\left(-I + (D^2w(y))^{-1}\right) = -F(D^2u(x)).$$

Suppose D^2u is diagonalized at x_0 . Then at x_0 we have

$$G^{ii} = F^{ii} \cdot w_{ii}^{-2} = F^{ii}(1 + u_{ii})^2. \quad (3.1)$$

Lemma 3.1. Let $n \geq 3$, $1 \leq l < k \leq n$ with $l = k-1, k-2$. Let $f \in C^{1,1}(B_{10} \times \mathbb{R})$ be a positive function and let $u \in C^4(B_{10})$ be a convex solution of (1.1). For any $x_0 \in B_9$, suppose that D^2u is diagonalized at x_0 such that $\lambda_1 \geq \dots \geq \lambda_n$.

Assume that Assumption 1.1 holds. Suppose $\lambda_1 \geq K_0$, where K_0 is the constant in Assumption 1.1. Let $b = \ln \lambda_1$, then at $y(x_0)$, we have

$$\sum_i G^{ii} b_{ii}^* \geq -C,$$

in the viscosity sense, where C depends only on $n, k, \|u\|_{C^{0,1}(\overline{B}_9)}, \min_{\overline{B}_9 \times [-M, M]} f$ and $\|f\|_{C^{1,1}(\overline{B}_9 \times [-M, M])}$. Here M is a large constant satisfying $\|u\|_{L^\infty(\overline{B}_9)} \leq M$. We use the notation $b^*(y) = b(x(y))$.

Proof. By the chain rule, we have

$$\frac{\partial b}{\partial x_i} = \sum_j \frac{\partial y^j}{\partial x^i} \frac{\partial b^*}{\partial y^j} = \sum_j (\delta_{ij} + u_{ij}) b_j^*. \quad (3.2)$$

Consequently,

$$\begin{aligned} \frac{\partial^2 b}{\partial x_i^2} &= \sum_j u_{ij} b_j^* + \sum_{j,p} (\delta_{ij} + u_{ij}) b_{jp}^* (\delta_{ip} + u_{ip}) \\ &= \sum_j u_{ij} b_j^* + (1 + u_{ii})^2 b_{ii}^*. \end{aligned}$$

Together with (3.1), we have

$$\begin{aligned} \sum_i F^{ii} b_{ii} &= \sum_{i,j} F^{ii} u_{ij} b_j^* + \sum_i F^{ii} (1 + u_{ii})^2 b_{ii}^* \\ &= \sum_j f_j b_j^* + \sum_i G^{ii} b_{ii}^*. \end{aligned}$$

Together with Lemma 2.5, we obtain

$$\sum_i G^{ii} b_{ii}^* = \sum_i F^{ii} b_{ii} - \sum_j f_j b_j^* \geq c(n, k) \sum_i F^{ii} b_i^2 - \sum_j f_j b_j^* - C,$$

where C is a constant depending only on $n, k, \|u\|_{C^{0,1}(\overline{B}_9)}, \min_{\overline{B}_9 \times [-M, M]} f$ and $\|f\|_{C^{1,1}(\overline{B}_9 \times [-M, M])}$. It may change from line to line.

By (3.1) and (3.2), we have

$$\sum_i F^{ii} b_i^2 = \sum_i G^{ii} (1 + u_{ii})^{-2} \left((1 + u_{ii}) b_i^* \right)^2 = \sum_i G^{ii} (b_i^*)^2.$$

It follows that

$$\sum_i G^{ii} b_{ii}^* \geq c(n, k) \sum_i G^{ii} (b_i^*)^2 - \sum_j f_j b_j^* - C \quad (3.3)$$

$$\begin{aligned}
&\geq c(n, k) \sum_i G^{ii} (b_i^*)^2 - C \sum_j |b_j^*| - C \\
&\geq -C \sum_i \frac{1}{G^{ii}} - C.
\end{aligned}$$

By Lemma 2.1 and Lemma 2.2, we have

$$\begin{aligned}
F^{ii} &= \frac{\sigma_k^{ii}}{\sigma_l} - \frac{\sigma_k \sigma_l^{ii}}{\sigma_l^2} = \frac{\sigma_{k-1}(|i) \sigma_l - \sigma_k \sigma_{l-1}(|i)}{\sigma_l^2} \\
&= \frac{\sigma_{k-1}(|i) (\lambda_i \sigma_{l-1}(|i) + \sigma_l(|i)) - (\lambda_i \sigma_{k-1}(|i) + \sigma_k(|i)) \sigma_{l-1}(|i)}{\sigma_l^2} \\
&= \frac{\sigma_{k-1}(|i) \sigma_l(|i) - \sigma_k(|i) \sigma_{l-1}(|i)}{\sigma_l^2}.
\end{aligned}$$

Since $l < k$, the Newton–Maclaurin inequality implies

$$\frac{1}{c(n, k)} \frac{\sigma_{k-1}(|i) \sigma_l(|i)}{\sigma_l^2} \leq F^{ii} \leq \frac{\sigma_{k-1}(|i) \sigma_l(|i)}{\sigma_l^2}, \quad (3.4)$$

where $c(n, k)$ is a positive constant depending only on n and k , and may change from line to line.

Case 1: $l = k - 1$.

For $1 \leq i < k$, we have

$$\sigma_{k-1}(|i)(1 + \lambda_i) \geq \frac{1}{c(n, k)} \frac{\sigma_k}{\lambda_i} \cdot \lambda_i \geq \frac{\sigma_k}{c(n, k)}.$$

For $k \leq i \leq n$, we have

$$\sigma_{k-1}(|i)(1 + \lambda_i) \geq \frac{\sigma_{k-1}}{c(n, k)}.$$

Together with (3.1) and (3.4), we obtain

$$G^{ii} \geq \frac{1}{C_0}, \quad (3.5)$$

where C_0 is a positive constant depending only on n , k , $\min_{\overline{B}_9 \times [-M, M]} f$ and $\max_{\overline{B}_9 \times [-M, M]} f$, and may change from line to line.

For $1 \leq i < k$, by Lemma 2.3 we have

$$\sigma_{k-1}(|i)(1 + \lambda_i) \leq C_0 \frac{\sigma_k}{\lambda_i} \cdot \lambda_i \leq C_0 \sigma_k.$$

For $k \leq i \leq n$, we have

$$\sigma_{k-1}(|i)(1 + \lambda_i) \leq C_0 \sigma_{k-1}.$$

Together with (3.1) and (3.4), we then have

$$G^{ii} \leq C_0. \quad (3.6)$$

Combining (3.5) and (3.6), we obtain

$$\frac{1}{C_0} \leq G^{ii} \leq C_0. \quad (3.7)$$

Substituting into (3.3), we conclude that

$$\sum_i G^{ii} b_{ii}^* \geq -C \sum_i \frac{1}{G^{ii}} - C \geq -C.$$

Case 2: $l = k - 2$.

For $1 \leq i < k - 1$, by Lemma 2.4 we have

$$\frac{\sigma_{k-1}(|i)\sigma_{k-2}(|i)}{\sigma_{k-2}^2}(1 + \lambda_i)^2 \geq \frac{1}{c(n, k)} \frac{\sigma_k \sigma_{k-1}}{\lambda_i^2 \sigma_{k-2}^2} \cdot \lambda_i^2 \geq \frac{\lambda_{k-1}}{C_0}.$$

For $i = k - 1$, again by Lemma 2.4 we have

$$\frac{\sigma_{k-1}(|i)\sigma_{k-2}(|i)}{\sigma_{k-2}^2}(1 + \lambda_i)^2 \geq \frac{1}{c(n, k)} \frac{\sigma_k}{\lambda_i \sigma_{k-2}} \cdot \lambda_i^2 \geq \frac{\lambda_{k-1}}{C_0}.$$

For $k \leq i \leq n$, Lemma 2.4 yields

$$\frac{\sigma_{k-1}(|i)\sigma_{k-2}(|i)}{\sigma_{k-2}^2}(1 + \lambda_i)^2 \geq \frac{1}{c(n, k)} \frac{\sigma_{k-1}}{\sigma_{k-2}} \geq \frac{\lambda_{k-1}}{c(n, k)}.$$

Together with (3.1) and (3.4), we obtain

$$G^{ii} \geq \frac{\lambda_{k-1}}{C_0}. \quad (3.8)$$

For $1 \leq i < k - 1$, Lemma 2.4 gives

$$\frac{\sigma_{k-1}(|i)\sigma_{k-2}(|i)}{\sigma_{k-2}^2}(1 + \lambda_i)^2 \leq C_0 \frac{\sigma_k \sigma_{k-1}}{\lambda_i^2 \sigma_{k-2}^2} \cdot \lambda_i^2 \leq C_0 \lambda_{k-1}.$$

For $i = k - 1$, Lemma 2.4 implies

$$\frac{\sigma_{k-1}(|i)\sigma_{k-2}(|i)}{\sigma_{k-2}^2}(1 + \lambda_i)^2 \leq C_0 \frac{\sigma_k}{\lambda_i \sigma_{k-2}} \cdot \lambda_i^2 \leq C_0 \lambda_{k-1}.$$

For $k \leq i \leq n$, Lemma 2.4 yields

$$\frac{\sigma_{k-1}(|i)\sigma_{k-2}(|i)}{\sigma_{k-2}^2}(1 + \lambda_i)^2 \leq C_0 \frac{\sigma_{k-1}}{\sigma_{k-2}} \leq C_0 \lambda_{k-1}.$$

Together with (3.1) and (3.4), we have

$$G^{ii} \leq \frac{\lambda_{k-1}}{C_0}. \quad (3.9)$$

Combining (3.8) and (3.9), we find

$$\frac{\lambda_{k-1}}{C_0} \leq G^{ii} \leq C_0 \lambda_{k-1}. \quad (3.10)$$

Together with Lemma 2.4, we obtain

$$-\sum_i \frac{1}{G^{ii}} \geq -\frac{C_0}{\lambda_{k-1}} \geq -C_0.$$

Substituting into (3.3), we conclude that

$$\sum_i G^{ii} b_{ii}^* \geq -C \sum_i \frac{1}{G^{ii}} - C \geq -C.$$

The lemma is proved. $\square$

We now state the mean value inequality.

Lemma 3.2. Let $n \geq 3$, $1 \leq l < k \leq n$ with $l = k-1, k-2$. Let $f \in C^{1,1}(B_{10} \times \mathbb{R})$ be a positive function and let $u \in C^4(B_{10})$ be a convex solution of (1.1).

Assume that Assumption 1.1 holds. Let

$$b = \ln \max\{\lambda_1, K_0\},$$

where λ_1 is the largest eigenvalue of D^2u , and K_0 is the constant in Assumption 1.1. Then we have

$$b(0) \leq C \int_{B_1} b(x) \sigma_{k-1}(D^2u(x)) dx + C,$$

where C depends only on $n, k, \|u\|_{C^{0,1}(\overline{B}_9)}, \min_{\overline{B}_9 \times [-M, M]} f$ and $\|f\|_{C^{1,1}(\overline{B}_9 \times [-M, M])}$. Here M is a large constant satisfying $\|u\|_{L^\infty(\overline{B}_9)} \leq M$.

Proof. Without loss of generality, we may assume $Du(0) = 0$, i.e. $y(0) = 0$.

By Lemma 2.3, Lemma 2.4, (3.7) and (3.10), we have

$$\sum_i G^{ii} \geq C_0,$$

where C_0 is a positive constant depending only on $n, k, \min_{\overline{B}_9 \times [-M, M]} f$ and $\max_{\overline{B}_9 \times [-M, M]} f$.

Since the maximum of two subsolutions is still a subsolution, Lemma 2.5 holds for b . Consequently, Lemma 3.1 holds as well. Therefore, we have

$$\sum_i G^{ii} (b^*(y) + A|y|^2)_{ii} \geq 0,$$

where A is a positive constant depending only on $n, k, \min_{\overline{B}_9 \times [-M, M]} f$ and $\max_{\overline{B}_9 \times [-M, M]} f$.

By (3.7), G^{ii} is uniformly elliptic for $l = k-1$. By (3.10), $\frac{G^{ii}}{\lambda_{k-1}}$ is uniformly elliptic for $l = k-2$. By the local maximum principle (Theorem 4.8 in [1]), we have

$$b^*(0) + A|0|^2 \leq C \int_{B_1^y} (b^*(y) + A|y|^2) dy.$$

Since $D^2u > 0$, the map $y(x) = Du(x) + x$ is uniformly monotone, i.e. $|y(x_I) - y(x_{II})| \geq |x_I - x_{II}|$, and hence $x(B_1^y) \subset B_1$. Together with the fact that $y(0) = 0$, we obtain

$$\begin{aligned} b(0) = b^*(0) &\leq C \int_{B_1} b(x) \det(D^2u(x) + I) dx + C \\ &\leq C \int_{B_1} b \cdot (\lambda_1 + 1) \cdots (\lambda_n + 1) dx + C \\ &\leq C \int_{B_1} b(1 + \sigma_1 + \cdots + \sigma_n) dx + C. \end{aligned}$$

Case 1: $l = k-1$.

By Lemma 2.3, $C^{-1} \leq \lambda_k \leq C$, thus

$$\sigma_{k-1} \geq C \sigma_{k-2} \geq \cdots \geq C \sigma_1.$$

Similarly,

$$\sigma_{k-1} \geq C \sigma_k \geq \cdots \geq C \sigma_n.$$

Thus

$$b(0) \leq C \int_{B_1} b \sigma_{k-1} dx + C.$$

Case 2: $l = k - 2$.

By Lemma 2.4, $C^{-1} \leq \lambda_{k-1} \lambda_k \leq C$, thus

$$\sigma_{k-1} \geq C \sigma_{k-2} \geq \cdots \geq C \sigma_1.$$

Similarly,

$$\sigma_{k-1} \geq C \sigma_k \geq \cdots \geq C \sigma_n.$$

Thus

$$b(0) \leq C \int_{B_1} b \sigma_{k-1} dx + C.$$

The lemma is proved. $\square$

4. Proof of Theorem 1.2. In this section, we complete the proof of Theorem 1.2 via integration by parts. Since F^{ij} is not divergence-free, we choose $H^{ij} = \sigma_l F^{ij}$ to perform the integration by parts, as in [13].

Define

$$H^{ij} = \sigma_l F^{ij} = \sigma_k^{ij} - \frac{\sigma_k \sigma_l^{ij}}{\sigma_l}. \quad (4.1)$$

Before proving the main theorem, we first establish an auxiliary lemma by induction. This lemma will help us to simplify the integration-by-parts argument.

Lemma 4.1. Let $n \geq 3$, $1 \leq l < k \leq n$ with $l = k - 1, k - 2$. Let $f \in C^{1,1}(B_{10} \times \mathbb{R})$ be a positive function and let $u \in C^4(B_{10})$ be a convex solution of (1.1). For any $x_0 \in B_9$, suppose that $D^2 u$ is diagonalized at x_0 . Let $b = \ln \lambda_1$, where λ_1 is the largest eigenvalue of $D^2 u$. Then at x_0 we have

$$\sum_i |b_i \sigma_j^{ii}| \leq C_0 \epsilon \sum_i H^{ii} b_i^2 + \frac{C_0}{\epsilon} \sigma_{k-1}, \quad \forall 1 \leq j \leq k - 1,$$

where ϵ is any positive constant and C_0 depends only on $n, k, \min_{\overline{B}_9 \times [-M, M]} f$ and $\max_{\overline{B}_9 \times [-M, M]} f$. Here M is a large constant satisfying $\|u\|_{L^\infty(\overline{B}_9)} \leq M$.

Proof. Without loss of generality, we may assume $\lambda_1 \geq \cdots \geq \lambda_n$.

By (4.1) and (3.4), we have

$$H^{ii} \geq \frac{1}{c(n, k)} \frac{\sigma_{k-1}(|i) \sigma_l(|i)}{\sigma_l}. \quad (4.2)$$

Case 1: $l = k - 1$.

For $1 \leq i < k$, by Lemma 2.3 and (4.2), we have

$$H^{ii} \geq \frac{1}{C_0} \frac{\sigma_k^2}{\lambda_i^2 \sigma_{k-1}} \geq \frac{1}{C_0} \frac{\sigma_{k-1}}{\lambda_i^2}.$$

For $k \leq i \leq n$, by Lemma 2.3 and (4.2), we have

$$H^{ii} \geq \frac{\sigma_{k-1}}{C_0}.$$

For $1 \leq i < j$, by Lemma 2.3 and the fact that $1 \leq j \leq k - 1$, we have

$$|b_i \sigma_j^{ii}| \leq C_0 |b_i| \frac{\sigma_j}{\lambda_i} \leq C_0 \epsilon \frac{\sigma_{k-1} b_i^2}{\lambda_i^2} + \frac{C_0}{\epsilon} \frac{\sigma_j^2}{\sigma_{k-1}} \leq C_0 \epsilon H^{ii} b_i^2 + \frac{C_0}{\epsilon} \sigma_j.$$

For $j \leq i < k$, by Lemma 2.3 and the fact that $1 \leq j \leq k-1$, we have

$$|b_i \sigma_j^{ii}| \leq C_0 |b_i| \sigma_{j-1} \leq C_0 \epsilon \frac{\sigma_{k-1} b_i^2}{\lambda_i^2} + \frac{C_0}{\epsilon} \frac{\sigma_{j-1}^2 \lambda_i^2}{\sigma_{k-1}} \leq C_0 \epsilon H^{ii} b_i^2 + \frac{C_0}{\epsilon} \sigma_j.$$

For $k \leq i \leq n$, by Lemma 2.3 and the fact that $1 \leq j \leq k-1$, we have

$$|b_i \sigma_j^{ii}| \leq C_0 |b_i| \sigma_{j-1} \leq C_0 \epsilon \sigma_{k-1} b_i^2 + \frac{C_0}{\epsilon} \frac{\sigma_{j-1}^2}{\sigma_{k-1}} \leq C_0 \epsilon H^{ii} b_i^2 + \frac{C_0}{\epsilon} \sigma_{j-1}.$$

Therefore,

$$\sum_i |b_i \sigma_j^{ii}| \leq C_0 \epsilon \sum_i H^{ii} b_i^2 + \frac{C_0}{\epsilon} \sigma_{k-1}, \quad \forall 1 \leq j \leq k-1.$$

Case 2: $l = k-2$.

For $1 \leq i < k-1$, by Lemma 2.4 and (4.2), we have

$$H^{ii} \geq \frac{1}{C_0} \frac{\sigma_k \sigma_{k-1}}{\lambda_i^2 \sigma_{k-2}} \geq \frac{1}{C_0} \frac{\sigma_{k-1}}{\lambda_i^2}.$$

For $i = k-1$, by Lemma 2.4 and (4.2), we have

$$H^{ii} \geq \frac{1}{C_0} \frac{\sigma_k \sigma_{k-2}}{\lambda_i \sigma_{k-2}} \geq \frac{1}{C_0} \frac{\sigma_{k-2}}{\lambda_i} \geq \frac{1}{C_0} \frac{\sigma_{k-1}}{\lambda_i^2}.$$

For $k \leq i \leq n$, by Lemma 2.4 and (4.2), we have

$$H^{ii} \geq \frac{\sigma_{k-1}}{C_0}.$$

For $1 \leq i < j$, by Lemma 2.4 and the fact that $1 \leq j \leq k-1$, we have

$$|b_i \sigma_j^{ii}| \leq C_0 |b_i| \frac{\sigma_j}{\lambda_i} \leq C_0 \epsilon \frac{\sigma_{k-1} b_i^2}{\lambda_i^2} + \frac{C_0}{\epsilon} \frac{\sigma_j^2}{\sigma_{k-1}} \leq C_0 \epsilon H^{ii} b_i^2 + \frac{C_0}{\epsilon} \sigma_j.$$

For $j \leq i < k$, by Lemma 2.4 and the fact that $1 \leq j \leq k-1$, we have

$$|b_i \sigma_j^{ii}| \leq C_0 |b_i| \sigma_{j-1} \leq C_0 \epsilon \frac{\sigma_{k-1} b_i^2}{\lambda_i^2} + \frac{C_0}{\epsilon} \frac{\sigma_{j-1}^2 \lambda_i^2}{\sigma_{k-1}} \leq C_0 \epsilon H^{ii} b_i^2 + \frac{C_0}{\epsilon} \sigma_j.$$

For $k \leq i \leq n$, by Lemma 2.4 and the fact that $1 \leq j \leq k-1$, we have

$$|b_i \sigma_j^{ii}| \leq C_0 |b_i| \sigma_{j-1} \leq C_0 \epsilon \sigma_{k-1} b_i^2 + \frac{C_0}{\epsilon} \frac{\sigma_{j-1}^2}{\sigma_{k-1}} \leq C_0 \epsilon H^{ii} b_i^2 + \frac{C_0}{\epsilon} \sigma_{j-1}.$$

Therefore,

$$\sum_i |b_i \sigma_j^{ii}| \leq C_0 \epsilon \sum_i H^{ii} b_i^2 + \frac{C_0}{\epsilon} \sigma_{k-1}, \quad \forall 1 \leq j \leq k-1.$$

The lemma is proved. $\square$

Proof of Theorem 1.2.

Proof. Let $b = \ln \max\{\lambda_1, K_0\}$ as in Lemma 3.2.

Let φ be a cutoff function such that $\varphi = 1$ on B_1 and $\varphi = 0$ outside B_2 . By Lemma 2.1 and the fact that $\sum_i (\sigma_{k-1}^{ij})_i = 0$, we have

$$\begin{aligned} \int_{B_1} b \sigma_{k-1} dx &\leq \int_{B_2} \varphi b \sigma_{k-1} dx = \frac{1}{k-1} \sum_{i,j} \int_{B_2} \varphi b \sigma_{k-1}^{ij} u_{ij} dx \\ &= -\frac{1}{k-1} \sum_{i,j} \int_{B_2} (\varphi_i b + \varphi b_i) \sigma_{k-1}^{ij} u_j dx \end{aligned}$$

$$\leq C \int_{B_2} b \sigma_{k-2} dx - \frac{1}{k-1} \sum_{i,j} \int_{B_2} \varphi b_i \sigma_{k-1}^{ij} u_j dx,$$

where C is a constant depending only on n, k and $\|u\|_{C^{0,1}(\overline{B_9})}$.

In what follows, we denote by C a constant depending only on $n, k, \|u\|_{C^{0,1}(\overline{B_9})}$, $\min_{\overline{B_9} \times [-M, M]} f$ and $\|f\|_{C^{1,1}(\overline{B_9} \times [-M, M])}$, which may change from line to line. Here M is a large constant satisfying $\|u\|_{L^\infty(\overline{B_9})} \leq M$.

Together with Lemma 3.2, we obtain

$$b(0) \leq C \int_{B_2} b \sigma_{k-2} dx - \frac{C}{k-1} \sum_{i,j} \int_{B_2} \varphi b_i \sigma_{k-1}^{ij} u_j dx + C. \quad (4.3)$$

By Lemma 4.1, we have

$$\left| -\frac{C}{k-1} \sum_{i,j} \int_{B_2} \varphi b_i \sigma_{k-1}^{ij} u_j dx \right| \leq C \sum_i \int_{B_2} H^{ij} b_i b_j dx + C \int_{B_2} \sigma_{k-1} dx.$$

It follows that

$$b(0) \leq C \int_{B_2} b \sigma_{k-2} dx + C \sum_i \int_{B_2} H^{ij} b_i b_j dx + C \int_{B_2} \sigma_{k-1} dx + C.$$

Using Lemma 4.1 repeatedly, we obtain

$$b(0) \leq C \sum_i \int_{B_3} H^{ij} b_i b_j dx + C \int_{B_3} \sigma_{k-1} dx + C. \quad (4.4)$$

Since the maximum of two subsolutions is still a subsolution, it follows from Lemma 2.5 that $b = \ln \max\{\lambda_1, K_0\}$ satisfies

$$\sum_{i,j} H^{ij} b_{ij} \geq c(n, k) \sum_{i,j} H^{ij} b_i b_j - C \sigma_{k-1},$$

in the viscosity sense.

By Theorem 1 in [10], b also satisfies the above inequality in the distribution sense. Let Φ be a cutoff function such that $\Phi = 1$ on B_3 and $\Phi = 0$ outside B_4 . Then

$$\begin{aligned} & \sum_{i,j} \int_{B_3} H^{ij} b_i b_j dx \leq \sum_{i,j} \int_{B_4} \Phi^2 H^{ij} b_i b_j dx \\ & \leq \frac{1}{c(n, k)} \sum_{i,j} \int_{B_4} \Phi^2 H^{ij} b_{ij} dx + C \int_{B_4} \sigma_{k-1} dx \\ & \leq -\frac{2}{c(n, k)} \sum_{i,j} \int_{B_4} \Phi \Phi_j H^{ij} b_i dx - C \sum_{i,j} \int_{B_4} \Phi^2 (H^{ij})_j b_i dx + C \int_{B_4} \sigma_{k-1} dx \\ & \leq \frac{1}{2} \sum_i \int_{B_4} \Phi^2 H^{ij} b_i b_j dx + C \sum_i \int_{B_4} \Phi_i \Phi_j H^{ij} dx + C \sum_{i,j} \int_{B_4} \Phi^2 f_j \sigma_l^{ij} b_i dx \\ & \quad + C \int_{B_4} \sigma_{k-1} dx. \end{aligned}$$

We used the fact that $\sum_j (H^{ij})_j = \sum_j \left((\sigma_k^{ij})_j - (f \sigma_l^{ij})_j \right) = -\sum_j f_j \sigma_l^{ij}$ in the last inequality.

Consequently,

$$\frac{1}{2} \sum_{i,j} \int_{B_4} \Phi^2 H^{ij} b_i b_j dx \leq C \sum_i \int_{B_4} \Phi_i \Phi_j H^{ij} dx + C \sum_i \int_{B_4} \Phi^2 f_j \sigma_l^{ij} b_i dx + C \int_{B_4} \sigma_{k-1} dx.$$

Together with Lemma 2.1, Lemma 4.1 and the identity $H^{ij} = \sigma_k^{ij} - f \sigma_l^{ij}$, we have

$$\begin{aligned} & \frac{1}{2} \sum_{i,j} \int_{B_4} \Phi^2 H^{ij} b_i b_j dx \\ & \leq C \sum_i \int_{B_4} \Phi_i \Phi_j H^{ij} dx + C \epsilon \sum_i \int_{B_4} \Phi^2 H^{ij} b_i b_j dx + \frac{C}{\epsilon} \int_{B_4} \sigma_{k-1} dx \\ & \leq C \int_{B_4} (C \sigma_{k-1} - C \sigma_{l-1}) dx + \frac{1}{4} \sum_i \int_{B_4} \Phi^2 H^{ij} b_i b_j dx + C \int_{B_4} \sigma_{k-1} dx \\ & \leq \frac{1}{4} \sum_{i,j} \int_{B_4} \Phi^2 H^{ij} b_i b_j dx + C \int_{B_4} \sigma_{k-1} dx. \end{aligned}$$

Consequently,

$$\frac{1}{4} \sum_{i,j} \int_{B_4} \Phi^2 H^{ij} b_i b_j dx \leq C \int_{B_4} \sigma_{k-1} dx.$$

Plugging this into (4.4), we get

$$b(0) \leq C \int_{B_4} \sigma_{k-1} dx + C.$$

Let Ψ be a cutoff function such that $\Psi = 1$ on B_4 and $\Psi = 0$ outside B_5 . Then

$$\int_{B_4} \sigma_{k-1} dx \leq \int_{B_5} \Psi \sigma_{k-1} dx = -\frac{1}{n-1} \sum_{i,j} \int_{B_5} \Psi_i \sigma_{k-1}^{ij} u_j dx \leq C \int_{B_5} \sigma_{k-2} dx.$$

Thus

$$b(0) \leq C \int_{B_5} \sigma_{k-2} dx + C.$$

Repeating this procedure, we obtain

$$b(0) \leq C \int_{B_6} \sigma_1 dx + C \leq C.$$

The theorem is proved. □

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